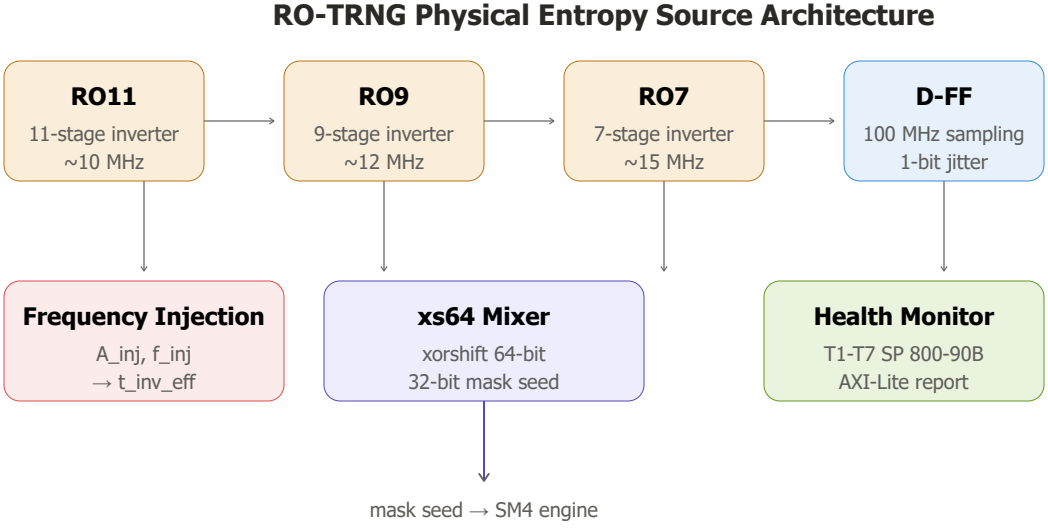
RO-TRNG Physical Entropy Source Architecture
RO11
11-stage inverter
~10 MHz
RO9
9-stage inverter
~12 MHz
RO7
7-stage inverter
~15 MHz
D-FF
100 MHz sampling
1-bit jitter
Frequency Injection
A_inj, f_inj
→ t_inv_eff
xs64 Mixer
xorshift 64-bit
32-bit mask seed
Health Monitor
T1-T7 SP 800-90B
AXI-Lite report
mask seed → SM4 engine